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# Chapter 1

# May

## 1.1 April 26

### 1.1.1 Nonlinear Beta

One of the main starting points is S. X. Chen, Gao, and Tang 2008 as I think it contains quite good review up till 2008. It's applied to diffusion models. For diffusion models also check out Hong and H. Li 2005 is about testing the diffusion model, I think it's based on Hardle and Mammen type of results, Aït-Sahalia, Fan, and Peng 2009 on the other hand, probably uses the Generalized Likelihood Ratio method(GLR).

Zu and Boswijk 2017 referenced the Hjellvik, Yao, and Tjøstheim 1998 which might the closest article to our purpose. It uses the Hardle and Mammen-type statistics, that is *integrated squared distance with smoothed parametric estimator.*

González-Manteiga and Crujeiras 2013 divides the methods into : smooth/kernel-based(like Hardle and Mammen), emprical regression process, and likelihood-ratio type. It says that the kernel based statistic can be derived as a **functional of emprical process/limiting Gaussian Process** which is interesting; or it can be derived using results in de Jong 1987.

### 1.1.2 Extension

Fan 2005 gives an overview of nonparametric methods in the stochastic dynamic asset pricing models. Especilly it talks about estimation of returns, volatility, state space

density.

GLR is related to Empirical Likelihood method, and EL is related to GMM, resulting in GEL (Newey and RJ Smith), should check it out later.

Perhaps should read some lecture notes on Hypothesis Testing.

## 1.2 April 27

For hypothesis testing problem. I looked at Lehnman TSH chapter 1. It's a long book and I think it's necessary to read through it, it doesn't have a lot of questions so probably a good way to exercise. The most important concepts are *size* $\alpha$ and *power* $\beta(\theta)$. Notice how power is a function of $\theta$ in the alternative.

González-Manteiga and Crujeiras 2013 says that the distribution of test statistics can be found via Empricial Process Method, so probably should check out the application part of van der Vaart and Wellner 1996.

Also, it seems that W. Newey 1985 and W. K. Newey 1985 are a focal point for parametric specification test. Similary we have Tauchen 1985.

From W. K. Newey 1985, we see that Halbert White has several papers on nonlinear regression and specification test in maximum likelihood framework. Should check out.

If we know the MSE of local linear estimator, we will know that it's consistent. And we are also close to asymptotic normality. Does it fit in the extremum estimator framework.

## 1.3 April 28

I think there is a relationship between Local Linear Regression and the GMM criterion. If we can put it together, we will have *at least* an estimate for each point $\hat{g}(x)$ that has correct asymptotic variance, it should also be consistent. Then we can apply likelihood ratio test to this estimate.

So need to figure out about local linear/polynomial regression and GLRT.

From Fan and Gijbels 1996: Chapter 2 gives an account of various smoothing methods in statistics, notice especially sections on wavelets and splines. Local polynomial

regression, assume that the second derivative exists. The univariate case is

$$\arg\min_{a,b} \sum \left(Y_j - a - b\left(x_0 - X_j\right)\right)^2 K_{h_n}\left(x_0 - X_j\right) \tag{1.1}$$

However the multivariate case is more complicated, perhaps we can consider the additive models. Check how it's done in Feng, Giglio, and Xiu 2017 and the network factor model.

Feng, Giglio, and Xiu 2017 indeed has a lot of factors and so perhaps even the simple Fama-Macbeth will fail in their settings, so it's a potential topic: how to extend the nonlinear test to higher-dimension.

Perhaps for the time being it's more reasonable to think about an additive model:

$$ER_t = \sum^{k} g_{j,t}(\beta_i) + u_t$$

Then we can test whether $\hat{g}$ is linear for the $j$-th factor.

Perhaps I should read Q. Li and Racine 2007, it has sections on multivariate local linear estimation and specification testing and also dependent data. I think it's probably the best introduction to nonparametrics. Also, first consider the parametric problem will be useful.

## 1.4 May 1, 2 and 3

I moved to London. Found one paper Horowitz and Spokoiny 2001, it's about uniform optial specification test, it has power against all directions of the alternative and thus superior in some direction. It's also mentioned in Q. Li and Racine 2007. Now I realise the key thing is to calculate the extra terms coming from time-series regression and see how important it is in the limiting distribution. It of course will depend on the relative size of $T$ and $N$

Can we derive it from the empirical process point of view.

Horowitz and Spokoiny 2001 tests the speicification with many different bandwidths $h \in H$ and then construct a uniformly consistent test. But it's very similar to Hardle and Mammen's.

What's the difference between the generalized likelihood ratio test and the

## 1.5 May 4

It's May the Fourth.

I read DWK Andrews's $\chi^2$ diagonstic statistic paper yesterday, it depends on sample splitting but it illustrates how empirical process method can be applied to derive the limiting distribution of statistics. It's different from the direct calculation of errors.

I browsed chapters in An Epsilon of Room. It is an interesting book that gives insights about the ideas, but be cautious that it could be misleading.

I did some calculation of the Specification test, it appears that I really need to get the limiting distribution of $\hat{\beta}$, $\hat{\gamma}$, which is in Asset Pricing:

### 1.5.1 Review of Chapters on GMM in Asset Pricing by Cochrane

Let $g_T(b) = \frac{1}{T}\sum_{t=1}^{T} u_t(b)$, then we estimate

$$\hat{b} = \arg\min g_T' W g_T$$

Let $S = E(gg')$, then $\text{var}(\beta)$ can be found via delta method, $g_T$, as it is a sample average, usually enjoys some CLT. Need to check Hall and Heyde 2014.

He also mentions some "general" GMM where

$$\hat{b} \,:\, a_T g_T(\hat{b}) = 0$$

notice that GMM is special case with $a_T = \frac{\partial g_T'}{\partial b} W$. And walks through the estimation and inference under this framework.

In Chapter 12, it discusses the Time Series, Cross-sectional and GMM estimation and testing of linear factor models.

For a single factor, the moments are:

$$g(b) = \begin{bmatrix} E(R_t - a - \beta f_t) \\ E[R_t - a - \beta f_t] f_t \\ E(R - \beta\gamma) \end{bmatrix} = 0$$

Let $\lambda$ be the selection vector of the cross-sectional part, we have, $a = \begin{bmatrix} I_{2N} & 0 \\ 0 & \lambda' \end{bmatrix}$.

$d = \frac{\partial g}{\partial b'}$, where $b = \begin{bmatrix} a' & \beta' & \gamma \end{bmatrix}'$.

When there are multiple factors, the moments are

$$\begin{bmatrix} I_N \otimes I_{K+1} & \\ & \lambda' \end{bmatrix} g(b) = 0$$

In the end, he compares the TS and CRS regression methods.

## 1.6 Chi-square diagnostic tests for econometric models: theory.

This is a review of DWK Andrews' Papers.

The take-away message is this: we have a weak convergence of a empirical process, we combine with continuous mapping theorem and the convergence in each component to arrive at the desired result.

Uses empirical process method to derive the distribution fo the test statistic.

$X_n^2(\hat{\Gamma}, \hat{\theta})$. We partition the region $Y \times X$ into $J$ disjoint cells. Let $D$ be the class of $J$partiitions of $Y \times X$. For each $n$, let $\hat{\Gamma}$ be a random element of $D$..

$$P_n(\gamma) = \frac{1}{n} \sum_{i=1}^{n} \gamma(Y_i, X_i)$$

where $\gamma$ is the vector of indicator functions of $(Y_i, X_i) \in \gamma_j$.

Then define

$$F_n(\gamma, \theta) = \frac{1}{n} \sum_{i=1}^{n} F(\gamma, X_i, \theta).$$

**Conditional Empirical process.**

$$v_n(\cdot, \theta) = \sqrt{n}(P_n(\cdot) - F_n(\cdot, \theta))$$

indexed by elements of $\gamma \in D$.

Good thing about chi-squared test, compared to Wald, LR and LM tests is that the latter has asymptotic optimality properties for a parametric subclass, while the chi-squared test has wider range of alternatives.

Pearson's chi-squared test was the first method devised for testing the appropriateness of a specified statistical model and it's quite important. But it uses an estimator called multinomial Maximum likelihood and its limiting distribution is a mixture of chi-squares.

This article reduces to several existing chi-square tests in the literature.

Partitioning method: Nonparametric partitioning, good for general Goodness-of-fit tests.

There are some examples of econometric models where the method can be applied. Should check on this later.

Condition: some smoothness condition on the conditional density $f(y \mid x, \theta)$ or the score $s(y \mid x, \theta)$. Also some linearization assumption.

## 1.7 May 5

Reread asset pricing book on the cross-sectional regression. Now read the Fama-Macbeth section.

Cochrane 2001 says that when $\varepsilon_{it}$ are correlated cross-sectionally but not serially, and when $\beta_{it} = \beta_i$(doesn't change over time), then the FM procedure is the same as a pooled OLS or a Cross-sectional regression of the time averages with *corrected* standard deviations.

It got me thinking, could we put everything in a GMM framework:

$$g(b) = \begin{bmatrix} \text{moments to estimate} \beta \\ \text{moments to estimate} \gamma \\ \text{Moments on the linearity} \end{bmatrix}$$

Assume we have for the 1st regression:

$$Y_{it} = a + \beta_i' F_t + \varepsilon_{it}$$

We estimate $\beta_i$ with time series regression:

$$E\left[Y_{it} - a - b'F_t\right] = 0$$
$$E\left[\left(Y_{it} - a - b'F_t\right)F_t\right] = 0$$

The sample analogue is

$$\hat{g}_{1,i}(\theta) = \frac{1}{T}\sum_{t=1}^{T}\begin{bmatrix}(Y_{it} - a - b'F_t)\\(Y_{it} - a - b'F_t)F_t\end{bmatrix} \quad : \quad (K+1)\times 1 \tag{1.2}$$

$$\hat{g}_1(\theta) = \begin{bmatrix}g_{1,1}\\ \vdots \\ g_{1,N}\end{bmatrix} \quad : \quad N(K+1)\times 1 \tag{1.3}$$

The 2nd step regression under the null is:

$$Y_{it} = \beta_i'\gamma_t + u_{it}$$

which translates into:

$$E\left[\left(Y_{it} - \beta_i'\gamma_t\right)\gamma_t\right] = 0$$

with sample analogue:

$$\hat{g}_{2,t}(\theta) = \frac{1}{N}\sum_{i=1}^{N}\left[\left(Y_{it} - \beta_i'\gamma_t\right)\beta i\right] \quad : \quad K\times 1$$

In order to do the two steps in one go, we combine the moment conditions:

$$\hat{g}(\theta) = \begin{bmatrix}g_1(\theta)\\ g_{2,t}(\theta)\end{bmatrix} \; : \; (N(K+1)+K)\times 1$$

See W. K. Newey and McFadden 1994 we have: for $G(\theta)$ being the uniform limit of $\frac{\mathrm{d}}{\mathrm{d}\theta}\hat{g}(\theta)$, we have

$$\sqrt{n}\left(\hat{\theta} - \theta\right) \rightsquigarrow N(0, (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1})$$

I think here we have different scaling factor (N,T) in place.

There is a huge revelation, I shouldn't have thought of the two regressions as separate, rather the error term in the second step really is just $\beta(F - E(F)) + \varepsilon$ under the null hypothesis, that induces the correction needed for the generated covariate problem.

Under the null hypothesis we have model:

$$R_{it} = \beta_i' F_t + \varepsilon_{it} \tag{1.4}$$

## 1.8 May 6

**Question 1.8.1.** *1. what's Shanken's result, is it root N or root T. 2. what's the good and bad about the several tests.*

### 1.8.1 Q1: A second look at Shanken

$$E_i = \gamma_0 + b_{i,1}\gamma_1 + b_{i2}\gamma_2 \tag{1.5}$$

$$R_{it} = a_i + b_{i1}F_{1t} + b_{i2}F_{2t} + \varepsilon_{it} \tag{1.6}$$

Write it in matrix form and we can take average of the second equation over time:

$$\bar{R} = [1_N \ : \ B]\bar{\Gamma} + \bar{\varepsilon}$$

where

$$\bar{\gamma}_1 = \gamma_1 + \bar{F}_1 - E(F_Z)$$

It assumes that $\varepsilon_t$ are iid **over time**. Let $\text{var}(\varepsilon_t \mid F) = \Sigma$. It is mentioned that to derive testable hypotheses it often requires more than just APT. Roll's critique, then about conditioning information.

It also talks about the relationship between two-pass and ML methods. It touches upon the case when we have fixed $T$ and $N \longrightarrow \infty$.

To answer the question: *Shanken is based on fixed N framework*, and our $\hat{\gamma}$ is $\sqrt{T}$ consistent. Is it still applicable if we want to test the nonlinearity using nonparametric method?

### 1.8.2 Q2: Comparison between differnet methods from Cochrane

This is from Cochrane 2001.

### 1.8.3 On Testing Asset Pricing Models

**Question 1.8.2.** *What's the literature on testing asset pricing models when $N, T \longrightarrow \infty$, probably need to check Campbell et al. 1997.*

First, let's do Campbell et al. 1997, Chapter 5 is about CAPM. Empirical tests of Sharpe-Lintner CAPM foces on the following things:

1. the inercept is 0
2. there is no other factor accounting for the cross-sectional variation of the $ER^e$.
3. Market risk premium $E(R^{e_m})$ is positive.

Narkowitz 1959 started CAPM. Then there is Sharpe-Lintner, then Black gives a version without the riskfree asset: with implication:

$$ER_i = \alpha_{im} + \beta_{im} ER_m \quad \text{and} \quad \alpha_{im} = E(R_{0,m})(1 - \beta_{im})$$

for some zero-beta asset.

## 1.9 May 7

Did some calculation on the test for nonlinear beta. The main results are here. Notice that if we model the asset pricing dynamic as

$$Y_{it} = \beta_i' F_t + \varepsilon_{it}$$

Then under this null, we'll have for $\gamma = E(F_t)$,

$$E_T Y_{it} = \beta_i' \gamma + E_T \varepsilon_{it} + \beta_i'(E_T F_t - E(F_t))$$

with $u_i = E_T \varepsilon_{it} + \beta_i'(E_T F_t - E(F_t))$ and conditional on $\beta_i$, it's zero-mean, so no exogeneity.

If we do Fama Macbeth, we are regressing for each $t$, so there are no $E_T$.

One of the major issue is that $u_i$ are correlated and the correlation is not specified.

## 1.10 May 8

I reread the Shanken paper, and now I see how he proves the central limit. Bascially,there is only one source of error coming from $\varepsilon$ and then the second step estimation error contains both $\varepsilon_{it}$ and the uncertainty on the $F_t$. It is really necessary to take a time-series average at some point.

Now the critical question is how to prove a central limit theorem for a cross-sectionally correlated sample.

So I read Hall and Hyde's Martingale Limit Theorem book. It's a very interesting book.

## 1.11 May 9

### 1.11.1 Overview of what I want to do this week

1. Find a solution for the cross-sectionally correlated error terms in the nonlinear beta problem.

2. Read and do exercise on Hall and Heyde 2014, van der Vaart and Wellner 1996. And more on measuret theory and functional analysis.

3. Read new papers.

## 1.12 May 11

Now I am reading Hardle and Mammen 1993 and see how we can actually apply the $L^2$ statistics.

### 1.12.1 Hardle and Mammen

**Assumption 1.** Some assumptions:

1. $X_i$ lies in a compact set in $\mathbb{R}^d$, and marginal density of $X$; $i > 0$.
2. $m(\cdot)$ and $f(\cdot)$ are twice continuously differentiabel, $\pi \in \mathcal{C}^1$.
3. $\delta_n(\cdot)$ is uniformly in $x, n$ bounded and $c_n = n^{-1/2}h^{-d/4}$
4. $\sigma^2(x)$ is bounded away from $0, \infty$.
5. $E\exp\{t\varepsilon_i\}$ is uniformly bounded in $i, n$ for $|t|$ small enough.

And the assumption on the parametric regression:

$$m_{\hat{\theta}}(\cdot) - m_{\theta_0}(\cdot) = \frac{1}{n}\sum_{i=1}^{n}\langle g(\cdot), \varepsilon_i h(X_i)|g(\cdot), \varepsilon_i h(X_i)\rangle + o_p\left((n\log n)^{\frac{-1}{2}}\right)$$

$$T_n = nh^{d/2}\int\left(\hat{m}_n(x) - \mathcal{K}_{h,n}m_{\hat{\theta}(x)}\right)^2\pi(x)$$

Then

$$T_n \rightsquigarrow N\left(b_h + \int(\mathcal{K}\delta_n)^2\pi, V\right)$$

Compare with Q. Li and Racine 2007 will actually give much resemblence.

### 1.12.2 On Central Limit Theorem for U statistic

With dependent random variable, there is Gao and Hong 2007. I think there are at least three papers that I need to read: Hall 1984, de Jong 1987 and the Gao and Hong 2007. They may all be based on some martingale result.

Kuersteiner and Prucha 2018 CLT for quadratic terms in a panel settings.

From Kuersteiner and Prucha 2018 I found the paper Gabaix and Koijen 2020, which seems very interesting.

## 1.13 May 12

Let's first derive the result for $L^2$ statistic when the error contains the first step estimation error.

I read the chapter 5 and 6 in CLM. I reread the email from Prof. Linton, it seems that actually it contains the basic framework for everything that I have spent a lot of effort to understand.

## 1.14 May 13

### 1.14.1 Literature Search

One question I am interested in is where we can apply IV method in Finance. It should be useful given that GMM is closely related to IV estimation. If we have a good question, we can use GIV perhaps.

The main theorem of the de Jong 1987 is that

**1.1 Theorem 2.1 in de Jong 1987.** *If $W_n = \sum_{i,j} W_{ij}$ with $W_{ij} = w_{ij}(X_i, X_j) + w_{ji}(X_j, X_i)$, $X_i$ are independent and $EW_{ij}^2 = \sigma_{ij}^2 < \infty$ is **clean**, that is*

$$E\left(W_{ij} \mid X_i\right) = 0 \quad a.s.$$

*with variance $\sigma(n)^2$, if*

1. $\sigma(n)^{-2} \max_i \sum_{j=1}^n \sigma_{ij}^2 \longrightarrow 0$ *as* $n \longrightarrow \infty$,
2. $\sigma(n)^{-4} EW(n)^4 \longrightarrow 3$ *as* $n \longrightarrow \infty$

$$\sigma(n)^{-1} W(n) \rightsquigarrow N(0,1)$$

◊

This is based on Martingale CLT.

### 1.14.2 On Nonlinear Beta

Wang 2003 tests with conditioning information nonparametrically, he has another paper Wang 2002 has more technical details.

And notice that once he get an error estimate $u$ he tests that $E(u \mid x) = 0$ by running a linear regression of $u$ on all the conditioning variables. He also uses some results about the generalized U-statistics.

I wonder how to put the CAPM in a SDF framework: see Cochrane's online course and does it matter to test the nonlinearity?

So we could put everything in a SDF framework where the pricing kernel is

$$m = \left(1 - b'f\right) \quad E(mR_i^e) = 0$$

In terms of CAPM, let $r$ be excess return of $R$,

$$E(r_i) = E(r_m)\frac{Er_m r_i}{E(r_m)^2} \tag{1.7}$$

$$m = (1 - br_m) \quad \text{where} \quad b = \frac{Er_m}{E(r_m)^2}$$
$$E(r_i - br_m r_i) = 0$$

Here $b$ is not the betas but rather the risk premiums. So in terms of linearity we should test whether the function $m$ is a linear function of $r_m$.

It seems that conditional CAPM tests takes $\beta$ as a fundamental characteristic of an asset and the risk premium can be changing, something like this:

$$ER_i = \gamma(z)\beta \quad \text{where} \quad z \quad \text{is conditioning variables}$$

but the relationship is still linear in $\beta$

For our test it seems that we are allowing:

$$ER_i = g(\beta) \quad \text{where} \quad g' = \gamma$$

But I was wrong. The conditional CAPM allows both $\beta$ and $\gamma$ to be conditional. I don't think that Wang 2003 properly takes care of the conditioning nature of the $\beta$'s.

I need to practice going forward and back between expected-return-beta model and SDF representation.

## 1.15 May 14

### 1.15.1 Conditional CAPM doesn't explain the size and value premium

Lewellen and Nagel 2006 says that the variation in $\beta$ and risk premium is too small to account for the the pricing errors. And develop a simple *short-window* test for the conditional CAPM and says that this method doesn't need to consider which conditioning information to include.

They did a short-window time series regression test with *portfolios*, and analysed the std.dv like Fama-MacBeth, which should be problematic, especially considering their estimates are noisy.

There is also Freyberger, Neuhierl, and Weber 2020, which should be relevant.

For the testing with the estimated errors:

1. Generalized liklihood ratio specialized to the regression model case and compare to Hardle and Mammen.

2. Look through the proof in Q. Li and Racine 2007 and see how to apply the CLT of quadratic forms.

I checked the proof in Fan, C. Zhang, and J. Zhang 2001 in the varying coefficient models, where the regression function is $A(U)$, we base the result on the $RSS_0$ and $RSS_1$ which then rely on the difference $A_0(u) - \hat{A}(u) = a_n + R_n$, then a lot of approximation takes place, plus the CLT in de Jong 1987 gives the final result.

## 1.16 May 15

We start with trying to find generalization of the de Jong 1987. Let's read Gao and Hong 2007.

Notice that in the end the test statistic we might use is

$$\sum_{i,j} \hat{u}_{it}\hat{u}_{jt}K_h\left(\hat{\beta}_i - \hat{\beta}_j\right)$$

I did some calculation of the test statistic

$$\sum_{i \neq j} \hat{u}_i \hat{u}_j$$

based on the result when we do full cross-sectional regression.

$$\hat{u}_i = \bar{\varepsilon}_i - \hat{\beta}_i \hat{A}_\beta \bar{\varepsilon} - \left\{ \left( \hat{\beta}_i - \beta_i \right) \bar{\gamma} - \hat{\beta}_i \hat{A}_\beta \left( \hat{\beta} - \beta \right) \bar{\gamma} \right\}$$

### 1.16.1 Time Varying Asset Pricing model

Gagliardini, Ossola, and Scaillet 2016 seems to answer most of my questions in a parametric way. Important.

There are some features of the model: $i \in [0, 1]$ and we have discrete time, there is also random sampling from $[0, 1]$ but I guess it's not essential. Then they develop a model with linear factor structure, derive testable implication, considered estimation, confidence interval and specification test, which is still based on the squared distance measure. It's a rather complete paper.

## 1.17 May 16

### 1.17.1 On regression with cross-sectional correlation

Donald WK Andrews 2005 which is referenced in Gagliardini, Ossola, and Scaillet 2016 considers regression model with cross-sectional correlation due to common shocks. He assumes that the error term $u_i$ in a cross-sectional regression are i.i.d conditional on some unobserved common shock $C$.

It can accommodate arbitrary type of dependence provided that the sampling scheme is random. Can we generalize it to the specification test settings?

There are more nonparametric testing in panel data model. For example, Lin, Q. Li, and Sun 2014, but they only allow time sereis correlation and they are applied to the whole panel, not the second step.

### 1.17.2 Some other readings

On the new Econometrica, there is a paper about social learning equlibria, it's not applicable to financial trading, but I think a description of the financial market using the Game Theory approac to account for the sallient features:

1. Asymmetric behaviour: leverage effect.
2. GARCH type clustering of volatitliy
3. Randomness in return , but with fat tail.
4. Bubble and crashes.
5. Granularity. Power distribution.

Ludvigson has a chapter in the Handbook of Economics for Finance, which mentions heterogenous agents. Might worth a look.

### 1.17.3 Gabaix and Koijen GIV

It seems that any weights that are not uniform will get rid of the common component, is there a better way to construct the IV? How does it relate to nonparametric IV models. It allows for screening using news and economic sense, can we make it scientific.

In section 2, without a loop, can construct given demand disturbance $y_{it}$ in $D_{it} = QS_i(1 + y_{it})$ and $S_i$ size, construct instrument

$$z_t = \sum S_i y_{it} - \sum \frac{1}{N} y_{it}$$

is a valid instrument because it get rid of the systematic/common component in $y$ that might be correlated to the shock $\varepsilon_t$ to the pricing equation:

$$p_t = \alpha y_{St} + \varepsilon_t$$

Then it talks about classical estimation of demand and supply elasticity using GIV.

Section 2.5 talks about the possibility that the idiosyncratic part of the shock, that is $u_{it}$ in $\lambda_i \eta_t + u_{it}$, will actually contain a part that comes from shock to the loading $\lambda$ of the common shock $\eta$.

In section 5, they discuss a spillver model of sovereign debt.

## 1.18 May 18

First article today, Ross 1976 on arbitrage theory. CAPM says that ex ante

$$E_i = \rho + \lambda b_i$$

It is an economic theory, with assumptions on the utitlity function, it derives the result starting from a factor pricing relationship, that the mean returns are approximately linear functions of factor loadings.

Chamberlain and Rothschild 1983 extends the result to approximate factor structure.

Then I had a look at Xiong Wei's literature, it seems that we should combine econometrics/statistics really with macro/finance. Otherwise we are fighting with air.

## 1.19 May 19

### 1.19.1 On Asset Pricing

Let's read Duffie Chapter 1 and then Chamberlain and Rothschild. From Chapter 1 we know about state-price pricing of assets. There are various ways to derive the state-price relationship. Next should read chapter 2 and then followed by the continuous-time part.

Bouchaud and Potters 2009 is a very well-written and concise review of RMT, must read. Guhr, Mueller-Groeling, and Weidenmueller 1998 talks about RMT in the context of Quantam Physics. It might also be interesting.

### 1.19.2 On Chinese Economic Growth and Local Governemnt

I read the starting of Xiong 2018 which talks about the local governments influence on the economic growth in China, see Inbox.

There are some questions and potential extensions:

1. He assumed that Labour is immobile, can we use the model to account for the concentration of labour in China.

2. Here the interest rate is exogenous, but in China, the finance part is a bit tricky, I think the government might be able to get financing more easily and this will amplify the leverage problem.

3. A qustion is that: given this structure, especially that $A_{it}$ differ across regions, can the central goverment allocate governors to maximize the productivity of some regions(high ability governors for high productivity region or for example, if a region is highly agricultural, send someone with that expertise), and will this create a further inequality among regions.

4. This article focuses on Infrastructure which has an immediate(perhaps not) effect on the output, but the government actually need to balance between short-term and long-term investments like education and health system, will this create an inequality of goverment investment.

5. Most importantly, for a central goverment, is there a better way to incentize the local governors?

## 1.20 May 20

I read chaper 2 of Duffie in the morning and watched a video of linear algebra. Should take a closer look at the error structure in the Gagliardini, Ossola, and Scaillet 2016.

Potential questions:

1. Local Linear regression estimator asymptotic properties under cross-sectional correlation(arbitrary).

2. Specification test when there is cross-sectional correlation.

But these are not BIG questions. Another question is to estimate system parameters by observing the system evolving. Will it reveal eventually all features?

Donald W.K. Andrews 1995 contains uniform consistency result for near-epoch non-id nonparametric regression when $Y, X$ are estimated, should be useful.

Jarrow and Protter 2019 explains that CEF(a pooled asset that has premium/discount market price than the constituents) premium and discounts can be explained by rational bubble theory. There also mentions a way to estimate bubble.

Kleibergen and Zhan 2019, mentions that the the quality of the measurement of the risk factor(here consumption) is important, also, for macro factor, data has low frequency, so limited $T$ and large $N$. Easily spurious factor. Two major tests are Fama-Macbeth and Hansen 1982 GMM.

So this paper extends the GRS test to GRS-FAR test,

## 1.21 May 22

Half-attended the seminar given by Prof. Xiong from Princeton today. But today and yesterday were mostly wasted.

It seems meaningless and meaningful at the same time. I wonder if Karl Marx has doubted himself.

## 1.22 Sunday, May 24

### 1.22.1 Overview of Things I want to do

1. Finish the Linear Algebra and Measure Theory by May 31
2. Preread some chapters in Duffie, get the idea of asset pricing like really understand it. read papers
3. Write a draft version and send it to Prof. Linton with arbitrary random cross-sectional correlation(but potentially only bounded eigenvalues). Notice that the time-varying beta doesn't seem to justify the CAPM. by May 25
4. Do the empirical part. by May 31
5. Discuss 1. GIV paper 2. Xiong's paper on Chinese Economy. arrange it this week.

In addition, there are some **leisure fun readings**:

1. Capital in the 21st Century
2. Art Today
3. Stein's Method for bounding the error in approximation of distributions.
4. Tong's Lecture notes on Dynamics and Relativity.

### 1.22.2 Dynamics and Relativity

**Inertial Reference Frame and Newton's First Law**

- *Particles* are items so small that we only care about its position $\mathbf{x}(t)$ in a *reference frame* usually choosen to be Cartesian, and the first Law of Newton is that there exists an *inertial reference frame* where

$$\dot{\mathbf{x}} := \frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} = 0 \quad \text{if} \quad \mathbf{F} = 0$$

- About the inertial framework, we have *Galilean group* which are equivalent inertial reference framework. Two inertial reference frames is closed under the following transformations:

  1. Translations: $\mathbf{x}' = \mathbf{x} + \mathbf{a}$.
  2. Rotations: $\mathbf{x}' = R\mathbf{x}$ where $R'R = 1$.
  3. Boosts: $\mathbf{x}' = \mathbf{x} + \mathbf{v}t$ for a constant velocity $\mathbf{v}$.

- The *principle of relativity* tells us the law of physics is the same for any inertial frame.

- Then we are left with the choice of time, there is a notion of *absolute time*. Where for any two inertial frame:

$$t' = t + t_0$$

**Newton's Second Law**

Let *momentum* be defined as

$$\mathbf{p} := m\dot{\mathbf{x}}$$

where $m$ is the *inertial mass* of the particle.

Then the *equation of motion*, or Newton's Second Law states that:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{p} = \frac{\mathrm{d}}{\mathrm{d}t}m\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \dot{\mathbf{x}})$$

Usually $m$ doesn't change over time so we have:

$$m\ddot{\mathbf{x}} = \mathbf{F}(\mathbf{x}, \dot{\mathbf{x}}, t)$$

Also $\mathbf{F}$ usually doesn't change over time too.

*Remark.* One important notion from this Law is that the equation of motion is a *second order differential equation* which requires in general two initial conditions. This notion will carry over to quantum physics.

*Remark.* The Newtonian laws are found inadequate when the particles are

1. Too fast: close to $c$, then we have *special relativity*.
2. Too small: even the concepts such as a trajectory of a particle is ill-defined and the new framework is called *quantum mechanics*. But *energy* and *momentum* will be carried over.
3. When we describe the forces at play with particles , then we need to talk about *fields*. The equations governing the dynamics of the fields are always second order differential equations.

In the end it's *quantum field theory* that is the current best theory for describing all the phenomena.

### 1.22.3 Random Cross-sectional Correlation

Two things first, to understand normal Local Linear Regression when the data are iid. Then understand OLS estimator when there is cross-sectional correlation.

### 1.22.4 Capital in 21st Century

From Malthus, Ricardo, Karl Marx to Kuznets and then there comes a revival of interest in economics about the inequality and distribution of income and wealth.

## 1.23 May 25

First I browsed the second chapter in Tong's notes on Dynamics and Relativity. Then I calculated the MSE for Local Linear Regression again. It seems that really there are a lot to be learnt. The book by Fan and Gijbels contain the information.

I think it's important to connect Linear Algebra with Vector Calculus. So let's do some reading on vector calculus, it's also useful for understanding mechanics.

### 1.23.1 Vector Calculus

See my notes.

### 1.23.2 Nonlinear Beta

In section 3.5 of Gagliardini, Ossola, and Scaillet 2016, they propose a specification test and mentions another paper Kan, Robotti, and Shanken 2013 which describes the asymptotic distribution of $R^2$ statistic and comparison between different asset pricing models' $R^2$.

## 1.24 May 26

Today two missions:

1. Finish the linear algebra text up to spectral theorems.
2. Do the local linear regression when there is arbitrary
3. *Organize the math analysis part

### 1.24.1 Linear Algebra

I have organized stuff up to inner product space. It's gonna take one more day for spectral theorems and two more days for linear operators on vector spaces over $\mathbb{R}$ and $\mathbb{C}$.

### 1.24.2 Draft version

I sent an email to Prof. Linton about the cross-sectional correlation problem in non-linear bet test.

### 1.24.3 An Introduction to Measure Theory 1.5

I read the section 1.5. Notice that the basic convergence theorems are in 1.4.5. So should repeat them afterwards.

But here is the relationship between several types of convergence. It's quite illuminating to think of the results in terms of step functions.

## 1.25 May 27

Morning, I read and think about stuff on convergence mode, differentiation theorems and linear algebra. Now I have some sense about the spectral theorems.

Next: polar decomposition and SVD.

In the afternoon I read Pollard's book about construction of measure on product space which is closely related to independence and conditioning. I have created a file containing these readings and the readings of Hall's Martingale limit theorem book. I hope that it can grow into a book about probability and statistics.

I read some sections in Capital in the 21st Century.

### 1.25.1 Sieve Empirical Likelihood

It contains the proof of an important lemma in Fan, C. Zhang, and J. Zhang 2001. It uses empirical process technique, so it needs a careful reading.

I also read Chapter 3 of Kosorok 2008, which talks about semiparametric efficiency, the result is that to construct an efficient estimator $T$ we need the following result:

$$\sqrt{n}(T - \theta(P)) = \sqrt{n}\mathbb{P}\tilde{\theta} + o_p(1)$$

where $\tilde{\theta}$ is in the tangent space(somewhat like a score).

## 1.26 May 28

Finish reading on SVD. The next two chapters on linear algebra are on general linear operators. Then read through the section 1.5 of Tao's Measure Book. The exercise that fast convergence of $L^1$ implies almost uniform convergence is very illuminating. Notice how uniform convergence differ from pointwise convergence.

There are potentially two violations of uniform convergence: that happens at infinity $f_n(x) = \{n \leq x \leq n+1\}$ and that happens locally $f_{nx} = \frac{1}{nx}\{x > 0\}$. The first one requires some "boundedness" condition to control, see Tao's version of Egorof theorem, where we have locally uniform convergence. The second one requires we knock out a small set $E$ with $m(E) < \varepsilon$.

## 1.27 May 29

### 1.27.1 Linear Algebra

I read the part about the decomposition of general linear operator on a complex vector space. The notion of generalized eigenvector is important. $E(\lambda, T) = N(T - \lambda I)$, $G(\lambda, T) = N(T - \lambda I)^j$ for some $j$. The problem of decomposition is to find enough eigenvectors that can form a basis. Notice an counterexample: $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. It has $\lambda_1 = \lambda_2 = 1$ but $\dim E(1, T) = 1$.

### 1.27.2 Pollard's Probability

Then I read the part of Pollard on product measure and Tonelli-Fubini theorems. The theorem is proved based on *generating class argument for class of functions* and the characterisation of integrals as unique **increasing linear functional on** $\mathcal{M}^+$ with **monotone convergence result**. Then the product measure is related to independence where again the independence uses generating class arguments a lot.

### 1.27.3 Citations of Andrews 2005

I have put the papers that I think are related which also cite Donald WK Andrews 2005 in a cluster.

#### Lee and Peter M. Robinson 2016

I start with Lee and Peter M. Robinson 2016 which seems the most relavent. This paper adopts the framework of P. M. Robinson 2011. When dealing with cross-sectional correlation, there are several ways:

1. Assume a unobserved common factor, like Andrews, Pesaran 2006 metrica and Bai 2009 metrica.
2. Spatial(economic) distance
3. Spatial mixing(which rules out strong dependence)
4. Possibly nonstationary linear process of disturbances with dependence in regressors expressed in terms of departure of joint densities from the product of marginals.

It has a good review of kernel and sieve estimation under dependence.

Their model is:

$$Y_i = m(X_i) + U_i \quad \text{where} \quad U_i = \sigma(X_i)e_i$$

#### P. M. Robinson 2011

It says that Andrews, arbitrarily strong form of dependence in disturbances but with the data then generated by random sampling which may not be plausible.

Assume the error part is linear process.

$$e_i = \sum_j^{\infty} a_{ij}\varepsilon_j \quad \text{for} \quad i \leq n, n = 1, 2, \ldots$$

Need to compare this with Andrews' result and also the standard result for kernel regression.

Maybe we can do something about generalized likelihood ratio test with this framework.

### 1.27.4 Dynamics and Relativity

I read the part about force in a potential field? Let's define $V(x)$ be the *potential energy*, then a kind of force that is called *conservative* is

$$F(\mathbf{x}, \dot{\mathbf{x}}) = -\frac{\mathrm{d}V(\mathbf{x})}{\mathrm{d}\mathbf{x}}$$

Here we can prove that $E = \frac{1}{2}m\dot{\mathbf{x}} + V(\mathbf{x})$ is a constant, or *conserved.* $T := \frac{1}{2}m\dot{\mathbf{x}}$ is called *kinetic energy.* Consider for example, the gravitational potential and the potential of a *Harmonic oscillator.*

With this type of conservation, we can simplify the second order differential equation to a first order differential equation and the solution is of *elliptical integral.*

## 1.28 May 30

### 1.28.1 Linear Algebra

I read the chapter on decomposition of Complex Linear Operators. The take-away messages are:

1. $G(\lambda, T)$ the generalized eigenvectors provide enough vectors for a basis. With which the operator $T$ has a block-upper-triangular matrix.

2. There exists one basis, such that $T$ has a Jordan form representation:

$$M = \operatorname{diag}(A_1, A_2, \dots, A_m) \quad \text{where} \quad A_j = \begin{bmatrix} \lambda_j & 1 & & & \\ & \lambda_j & 1 & & \\ & & \lambda_j & 1 & \\ & & & \ddots & \ddots \end{bmatrix}$$

## 1.29 May 31

I browsed through the chapter on real linear operator on general vector spaces and I think the better route is to consider the determinant and the characteristic polynomials. The results are nevertheless important. Notice how complexification of operator is used.

# Chapter 2

# June

## 2.1 June Plan

Organize the following results:

1. Linear Algebra
2. Measure Theory
3. Specification tests
4. U-statistics
5. Empirical Process
6. Test of asset pricing models
7. On parametric estimation and hypothesis testing
8. On cross-sectional correlation

Do the following:

1. Matrix analysis
2. Functional Analysis
3. Martingale and stochastic processes
4. Empirical Process and martingale limit theorem

## 2.2 June 1

### 2.2.1 Linear algebra

Here notice the difference between *characteristic polynomial* and *minimal polynomial*:

$$\begin{aligned}\chi(A,\lambda) &= \det\{A-\lambda I\}\\ m(A,\lambda) &= \quad \text{the smallest monic polynomial such that} \quad m(A)=0\end{aligned}$$

which encodes the difference between an eigenspace and *generalized eigenspace.*

### 2.2.2 Measure Theory

I think they can all be written in a set theoretical form. Then there are some useful rules:

**2.1 Proposition (Set Proof).** *Let $S(x)$ be some statement given some parameters $x \in \mathcal{X}$, where $\mathcal{X}$ is the parameter space, that is*

$$\begin{aligned}S : \mathcal{X} &\longrightarrow \{1,0\}\\ x &\longmapsto S(x)\end{aligned}$$

*A **condition** $X$ is a subset of $\mathcal{X}$. We define the union and intersection as, for $x_0 \in X$,*

$$\begin{aligned}S(x_0) \le \bigcup_{x\in X} S(x) &= \sup_{x\in X} S(x)\\ \bigcap_{x\in X} S(x) &= \prod_{x\in X} S(x) \le S(x_0)\end{aligned}$$

1. *$\bigcup_x \bigcup_y S(x,y) = \bigcup_y \bigcup_x S(x,y)$ and $\bigcap_x \bigcap_y S(x,y) = \bigcap_y \bigcap_x S(x,y)$.*
2. *$\bigcup_{y\in Y} \bigcap_{x\in X} S(x,y) \le \bigcap_{x\in X} \bigcup_{y\in Y} S(x,y)$. In the second expression, we can choose $y = y(x)$.*

*If for some $X_0 \subset X$, $\bigcap_{x\in X_0} S(x) = 1$ then $X_0 \subset \{x : S(x) = 1\} \subset X$.*

Now put it in use, we prove that uniform convergence $\implies$ pointwise convergence.

**Convergence Modes.** Let $f_n, f : X \longrightarrow \mathbb{C}$, the dependence of statement function $S$ on $(f_n), f$ are omitted:

1. We say $f_n \rightrightarrows f$ on $E \subset X$ if

$$\bigcap_{\varepsilon>0} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \bigcap_{x\in E} \mathbf{1}(|f_n(x) - f(x)| < \varepsilon) = 1$$

2. Pointwise convergence $f_n \longrightarrow f$ for all $x \in E \subset X$ if

$$\bigcap_{\varepsilon} \bigcap_{x\in E} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \mathbf{1}(|f_n(x) - f(x)| < \varepsilon) = 1$$

3. Almost every convergence: $f_n \longrightarrow f$, *a.e.* if there exists a null set $E \subset X$ such that

$$\bigcap_{\varepsilon} \bigcap_{x\in E^c} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \mathbf{1}(|f_n(x) - f(x)| < \varepsilon) = 1$$

4. *Almost uniform convergence*, $f_n \rightrightarrows_a f$ if $(X, \mathcal{F}, \mu)$ is a measure space,

$$\bigcap_{\delta>0} \bigcup_{E\in\mathcal{F},\mu(E)<\delta} \bigcap_{\varepsilon} \bigcup_{N} \bigcap_{n\geq N} \bigcap_{x\in E^c} \mathbf{1}(|f_n(x) - f(x)| < \varepsilon) = 1$$

*Remark.* Notice that in uniform convergence, $N$ depends only on $\varepsilon$ whereas in pointwise, it can depend on $\varepsilon$ and $x$.

**Example.** Now prove that $\forall x \in X, f_n \longrightarrow f$ implies $f_n \longrightarrow f\, a.e.{-}X$ which means(assume that $X$ is a complete measure space),

$$\bigcup_{E_1\in\{E\subset X : \mu(E)=0\}} \bigcap_{x\in E_1^c} \bigcap_{\varepsilon} \bigcup_{N} \bigcap_{n\geq N} S(n, \varepsilon, x) = 1$$

*Proof.* Let $S(E) = \bigcap_{x\in E^c} \bigcap_{\varepsilon} \bigcup_N \bigcap_{n\geq N} \mathbf{1}(|f_n(x) - f(x)| < \varepsilon)$, then $S(\emptyset) = 1$,

$$1 = S(\emptyset) \leq \bigcup_{E\in\text{null sets}} S(E)$$

And we are done. □

*Remark.* The important thing is to choose $S(\cdot)$ wisely.

**2.2 Egoroff's Theorem.** *Let $X$ be a probability space, then, pointwise $f_n \longrightarrow f$ implies that $f_n \rightrightarrows_a f$.* ◊

*Proof.* WLOG, $f = 0$. We have $\bigcap_\varepsilon \bigcap_{x\in X} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \mathbf{1}(|f_n(x)| < \varepsilon) = 1$, we understand that this statement is true for any value of $(\varepsilon, x, n)$ that satisfies the parameter constraints($\varepsilon > 0, x \in X, n \geq N(\varepsilon, x)$).

Now we want to move $\bigcap_x$ to the right by modifying the parameter space, this trick is important. Fix $\delta > 0$, we want construct a $E = E(\delta) \in \mathcal{F}$ satisfying $\mu(E) < \delta$ and

$$\bigcap_{\varepsilon>0} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \bigcap_{x\in E^c} \mathbf{1}(|f_n(x)| < \varepsilon) = 1 \quad \text{a statement depending on} \quad E$$

From pointwise convergence we know $X = S_{p,x}^{-1}(1)$, where $p$ stands for pointwise,

$$S_{p,x} = \bigcap_{\varepsilon} \bigcup_{N\in\mathbb{N}} \bigcap_{n\geq N} \mathbf{1}(|f_n(x)| < \varepsilon) = \bigcap_{k\in\mathbb{N}} \bigcup_{N} \bigcap_{n\geq N} \mathbf{1}\left(|f_n| < \frac{1}{k}\right).$$

So $S_{p,x}^c = \bigcup_k \bigcap_N \bigcup_{n\geq N} \mathbf{1}\left(|f_n| \geq \frac{1}{k}\right)$, $\left(S_{p,x}^c\right)^{-1}(1) = \emptyset$ □

After a day of calculating, I think it's not always(if not seldom) easy to write things in the set form. Perhaps, it's easier to consider everything as a function.

**2.3 Proposition (Convergence Mode).** *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $f_n, f : X \longrightarrow \mathbb{C}$. Then:*

1. *$f_n \longrightarrow f$ is the usual pointwise convergence. $f_n \rightrightarrows f$ is uniform convergence.*
2. *$f_n \longrightarrow f$, a.e. if....*

## 2.3 Jun 2

### 2.3.1 Linear Algebra

I did the proof of inner product space up to the projection. Next, Riesz representation and adjoints. With these tools, we can get the spectral theorems.

Then there are polar and singular value decomposition.

### 2.3.2 Conditioning and Martingale

**P. M. Robinson 2011**

I read the consistency part. The bounding is mostly based on $L^2$.

Asymptotic normality is always harder. But I need to see it done.

At this point I think it's imperative to go through Mathematical Statistics and see what really are the procedures. And maybe organize the parametric part.

## 2.4 Jun 3

I read SLLN via blocking in Pollard and mathematical statistics.

## 2.5 Jun 4

I read on the hypothesis testing part.

Notice that P. M. Robinson 2011 only gave a consistency and CLT for kernel regression estimator, and the limiting variance depends on which part is larger: the cross-sectional correlation or the time variance. Hence it's not complete.

Donald WK Andrews 2005 looks at random sampling design. I wonder if it can be relaxed such that conditional on $\mathcal{C}$, the random variables are not independent but the dependence are so small that they can be ignored.

Lee and Peter M. Robinson 2016 considers a HAC inference where they use studentization that doesn't require specification of economic distance. I wonder if Generalized Likelihood Ratio statitstics automatically does that.

### 2.5.1 Linear Algebra

I thought about the spectral theorems again and now I think I realize why normal and self-adjoint operators are good. Assume we have an eigenvector $v$ of $T$, then we want to show that $\{v\}^\perp$ is invariant under $T$, for which we need $\langle Tx|v\rangle$ for $x \perp v$, then we need somehow carry $T$ over to the other side. For normal operators, $T$ and $T^*$ share the same eigenvectors. For real inner product space, we need self-adjoint operator to guarantee that we have an eigenvector to begin with.

## 2.6 Jun 8

Three questions:

1. Under the framework of game theory, can we explain what's happening in Chinese economy?
2. Can we infer the ratio of uninformed investors in the market?
3. What game system could generate a Brownian Motion?

Some papers to read: Chang, S. X. Chen, and X. Chen 2015(with the papers reviewed in the first paragraph), Granular origin of aggregate fluctuation and GIV.

### 2.6.1 Linear Algebra

Thought about Jordan Basis Form. It is starting to make sense now.

### 2.6.2 Math Analysis

About the definition and least upper bound property of the real numbers.

### 2.6.3 Local Linear Regression

See Fan and Gijbels 1996 for the MSE and asymptotic normality of the local linear/polynomial regression methods in Chapter 3 and Chapter 6(resp. independent and dependent with mixing conditions).

The Econometrics of High Frequency Data review is in a tutorial collection of papers in a series, it might be worth checking out.

## 2.7 Jun 9

### 2.7.1 Nonlinear Beta: Estimation Using Local Linear

Now a question comes, how to estimate under the alternative. Maybe conditional CAPM will be helpful.

## 2.8 Jun 10

### 2.8.1 Math Analysis

About Sequence and Series.

### 2.8.2 About Matrix Analysis

Eigenvalue from optimization problem and polynomial.

### 2.8.3 Game Theory and Asset Pricing

There is a paper showing that actually *technical analysis traders* can make money and that is in violation of the fundamentalists. There are ways to generate volatility clustering.

## 2.9 Jun 11

I read Functional Analysis and Conditional Expectation.

Several things:

1. Properties of Conditional Expectations
2. Martingale Convergence Theorems
3. Macro Paper search
4. Generalized Likelihood Ratio

## 2.10 Jun 14

### 2.10.1 Matrix Analysis

About variational characterisation of eigenvalues of Hermitian matrices. About Weyl and Cauchy Interlacing theorems and consequences.

### 2.10.2 Macro

Z. Chen, He, and Liu 2020 who cites Xiong's paper is about the surge of shadow banking following Chinese 2009 stimulus.

1994 Tax-sharing reform- Local Govern(LG) can't borrow by themselves, so through LGFV(financing vehicles), which is state-owned enterprise (SOE), LG is the major shareholder of the SOE. Shifting from bank loans to nonbank loans.

NAO 2010 2013, survey. Interesting for knowing about Chinese shadow banking and local government debt.

### 2.10.3 A few things

It's interesting to consider how a complex system could evolve. There seems to be multple possibilities including: a unique equilibrium, multiple equilibria but stable, no equilibrium and everything in chaos, and a lot of possible states in between. It seems that in financical markets, things are usually somewhat in a stable equilibrium until some huge news totally disturbs the system and then it reverts back to a new equilibrium. The thing is it doesn't happen instanteneously.

It's also important to look at real data. In fact, from now on, each day I will devote at least

1. 30 minutes for matrix analysis
2. 30 minutes for real analysis
3. 30 minutes for probability
4. 30 minutes for real data: collection and analysis
5. 30 minutes for reading literature of the day.

## 2.11 Jun 15

Assume there are some ingredients:

1. Individual no arbitrage but the price might be different from a true fundamental value: everyone has a different belief about what the value of a stock/index is:

$x_i$, if the current price has difference more than trasaction cost $c$ then the agent will want to trade.

2. Whether they can trade or not depend on the position they have: i.e., *short is very costly* so in practice, we have no shorts.

3. Also they could have limited access to the market, so cross-sectionally, there are people who can more easily sell/buy stocks(they have more frequent chances to form a market),

Things we want to explain: leverage effect, volatility clustering, fat tail of returns, crashes.

## 2.12 Wednesday, 17 June 2020

GLR hasn't been extended to the case with dependent data. Escanciano, Pardo-Fernández, and Keilegom 2018 instead.

## 2.13 Thursday, 18 June 2020

- Definition of norms and inner product. Cauchy-Schwarz inequality.

- Algebraic properties of norm: If we have $m$ norms and a norm on $\mathbb{R}^m$ then the composition is also a norm.

- Analytic properties: equivalence of norms(generalized). Continuous functions on unit sphere(which is always compact in a finite-dimensional space) is going to be bounded.

  Definition: Dual Norm of a Pre-Norm and extended CS inequality. This extended CS inequlaity can be further extended to Holder inequality.

- Geometric Properties. Duality theorem.

For matrix, a *matrix norm* is a norm such that $\|AB\| \leq \|A\|\|B\|$. $l^p$ and Operator norm.

**2.4 Inverse of $A + e$.** *Let $A_n \longrightarrow A$ be a square matrix. Let $e_n$ be an error matrix, in that for some matrix norm $\|e_n\| = o_p(1)$. Then for a nonsingular matrix $A$, eventually $A + e$ is invertible and*

$$(A_n + e_n)^{-1} = \left[A_n\left(I + A_n^{-1}\right)e_n\right]^{-1} \longrightarrow A^{-1} + o_p(1)$$

◊

## 2.14 Monday, 22 June 2020

Today, I calculated the local linear regression for nonlinea beta and compare with the paper Escanciano, Pardo-Fernández, and Keilegom 2018. It seems that the steps can be broken down and it just requires some more effort.

Also I looked at Matrix Analysis Book. I need to continue on the three math projects and do literature search on

1. Emprical Tests on Asset Pricing Models.
2. Asset Pricing Theory.
3. Chinese Economy.

## 2.15 Tuesday, 23 June 2020

### 2.15.1 Matrix Analysis

We learned about *characteristic polynomial* which is

$$p_A(t) = \det(tI - A)$$

Then we show that the zeros of $p_A$ are the eigenvalues of $A$ together with multiplicity. Then as an example, we show the Brauer's Theorem which shows how to solve for the eigenvalues of $A + xy^*$ where $x$ is an eigenvector of $A$. It uses the property of determinant.

Then we discuss principal minors and elementary symmetric function.

# Bibliography


Aït-Sahalia, Yacine, Jianqing Fan, and Heng Peng (Sept. 1, 2009). "Nonparametric Transition-Based Tests for Jump Diffusions". In: *Journal of the American Statistical Association* 104.487, pp. 1102–1116. ISSN: 0162-1459. DOI: 10.1198/jasa.2009.tm08198. URL: https://doi.org/10.1198/jasa.2009.tm08198 (visited on 04/26/2020).

Andrews, Donald W.K. (June 1995). "Nonparametric Kernel Estimation for Semiparametric Models". In: *Econometric Theory* 11.3, pp. 560–586. ISSN: 0266-4666, 1469-4360. DOI: 10.1017/S0266466600009427. URL: https://www.cambridge.org/core/product/identifier/S0266466600009427/type/journal article (visited on 05/20/2020).

— (2005). "Cross-Section Regression with Common Shocks". In: *Econometrica* 73.5, pp. 1551–1585.

Bouchaud, J. P. and M. Potters (Oct. 7, 2009). "Financial Applications of Random Matrix Theory: A Short Review". In: arXiv: 0910.1205 [cond-mat, q-fin]. URL: http://arxiv.org/abs/0910.1205 (visited on 05/19/2020).

Campbell, John Y. et al. (1997). *The Econometrics of Financial Markets*. princeton University press.

Chamberlain, Gary and Michael Rothschild (1983). "Arbitrage, Factor Structure, and Mean-Variance Analysis on Large Asset Markets". In: *Econometrica* 51.5, pp. 1281–1304. ISSN: 0012-9682. DOI: 10.2307/1912275. JSTOR: 1912275.

Chang, Jinyuan, Song Xi Chen, and Xiaohong Chen (Mar. 1, 2015). "High Dimensional Generalized Empirical Likelihood for Moment Restrictions with Dependent Data". In: *Journal of Econometrics* 185.1, pp. 283–304. ISSN: 0304-4076. DOI: 10.1016/j.jeconom.2014.10.011. URL: http://www.sciencedirect.com/science/article/pii/S0304407614002553 (visited on 04/19/2020).

Chen, Song Xi, Jiti Gao, and Cheng Yong Tang (Feb. 2008). "A Test for Model Specification of Diffusion Processes". In: *The Annals of Statistics* 36.1, pp. 167–198. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/009053607000000659. URL: https://projecteuclid.org/euclid.aos/1201877298 (visited on 04/14/2020).

Chen, Zhuo, Zhiguo He, and Chun Liu (July 1, 2020). "The Financing of Local Government in China: Stimulus Loan Wanes and Shadow Banking Waxes". In: *Journal of Financial Economics* 137.1, pp. 42–71. ISSN: 0304-405X. DOI: 10.1016/j.jfineco.2019.07.009. URL: http://www.sciencedirect.com/science/article/pii/S0304405X20300179 (visited on 06/14/2020).

Cochrane, John Howland (2001). *Asset Pricing*. Princeton University Press. 556 pp. ISBN: 978-0-691-07498-6. Google Books: WpmtKSoM4KcC.

De Jong, Peter (1987). "A Central Limit Theorem for Generalized Quadratic Forms". In: *Probability Theory and Related Fields* 75.2, pp. 261–277.

Escanciano, Juan Carlos, Juan Carlos Pardo-Fernández, and Ingrid Van Keilegom (June 2018). "Asymptotic Distribution-Free Tests for Semiparametric Regressions with Dependent Data". In: *Annals of Statistics* 46.3, pp. 1167–1196. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/17-AOS1581. URL: https://projecteuclid.org/euclid.aos/1525313079 (visited on 06/17/2020).

Fan, Jianqing (2005). "A Selective Overview of Nonparametric Methods in Financial Econometrics". In: *Statistical Science*, pp. 317–337.

Fan, Jianqing and Irene Gijbels (Mar. 1996). *Local Polynomial Modelling and Its Applications: Monographs on Statistics and Applied Probability 66*. CRC Press. ISBN: 978-0-412-98321-4.

Fan, Jianqing, Chunming Zhang, and Jian Zhang (2001). "Generalized Likelihood Ratio Statistics and Wilks Phenomenon". In: *The Annals of Statistics* 29.1, pp. 153–193. ISSN: 0090-5364. JSTOR: 2674021.

Feng, Guanhao, Stefano Giglio, and Dacheng Xiu (2017). "Taming the Factor Zoo". In:

Freyberger, Joachim, Andreas Neuhierl, and Michael Weber (May 1, 2020). "Dissecting Characteristics Nonparametrically". In: *The Review of Financial Studies* 33.5, pp. 2326–2377. ISSN: 0893-9454. DOI: 10.1093/rfs/hhz123. URL: https://academic.oup.com/rfs/article/33/5/2326/5821383 (visited on 05/14/2020).

Gabaix, Xavier and Ralph SJ Koijen (2020). "Granular Instrumental Variables". In: *Available at SSRN 3368612*.

Gagliardini, Patrick, Elisa Ossola, and Olivier Scaillet (May 1, 2016). "Time-Varying Risk Premium in Large Cross-Sectional Equity Data Sets". In: *Econometrica* 84.3, pp. 985–1046. ISSN: 1468-0262. DOI: 10.3982/ECTA11069. URL: http://onlinelibrary.wiley.com/doi/10.3982/ECTA11069/abstract (visited on 01/25/2018).

Gao, Jiti and Yongmiao Hong (2007). "Central Limit Theorems for Generalized U –Statistics with Applications in Econometric Specification". In: p. 21.

González-Manteiga, Wenceslao and Rosa M. Crujeiras (2013). "An Updated Review of Goodness-of-Fit Tests for Regression Models". In: *Test* 22.3, pp. 361–411.

Guhr, Thomas, Axel Mueller-Groeling, and Hans A. Weidenmueller (June 1998). "Random Matrix Theories in Quantum Physics: Common Concepts". In: *Physics Reports* 299.4-6, pp. 189–425. ISSN: 03701573. DOI: 10.1016/S0370-1573(97)00088-4. arXiv: cond-mat/9707301. URL: http://arxiv.org/abs/cond-mat/9707301 (visited on 05/19/2020).

Hall, Peter (1984). "Central Limit Theorem for Integrated Square Error of Multivariate Nonparametric Density Estimators". In: *Journal of multivariate analysis* 14.1, pp. 1–16.

Hall, Peter and Christopher C. Heyde (2014). *Martingale Limit Theory and Its Application*. Academic press.

Hardle, W. and E. Mammen (Dec. 1993). "Comparing Nonparametric Versus Parametric Regression Fits". In: *The Annals of Statistics* 21.4, pp. 1926–1947. ISSN: 0090-5364, 2168-8966. DOI: 10.1214/aos/1176349403. URL: https://projecteuclid.org/euclid.aos/1176349403 (visited on 02/22/2020).

Hjellvik, Vidar, Qiwei Yao, and Dag Tjøstheim (1998). "Linearity Testing Using Local Polynomial Approximation". In: *Journal of Statistical Planning and Inference* 68.2, pp. 295–321.

Hong, Yongmiao and Haitao Li (2005). "Nonparametric Specification Testing for Continuous-Time Models with Applications to Term Structure of Interest Rates". In: *The Review of Financial Studies* 18.1, pp. 37–84.

Horowitz, Joel L. and Vladimir G. Spokoiny (2001). "An Adaptive, Rate-Optimal Test of a Parametric Mean-Regression Model Against a Nonparametric Alternative". In: *Econometrica* 69.3, pp. 599–631. ISSN: 1468-0262. DOI: 10.1111/1468-0262.00207. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1468-0262.00207 (visited on 05/02/2020).

Jarrow, Robert and Philip Protter (2019). "A Rational Asset Pricing Model for Premiums and Discounts on Closed-End Funds: The Bubble Theory". In: *Mathematical Finance* 29.4, pp. 1157–1170. ISSN: 1467-9965. DOI: 10.1111/mafi.12207. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/mafi.12207 (visited on 05/20/2020).

Kan, Raymond, Cesare Robotti, and Jay Shanken (2013). "Pricing Model Performance and the Two-Pass Cross-Sectional Regression Methodology". In: *The Journal of Finance* 68.6, pp. 2617–2649.

Kleibergen, Frank and Zhaoguo Zhan (2019). "Robust Inference for Consumption-Based Asset Pricing". In: *The Journal of Finance* 0 (ja). ISSN: 1540-6261. DOI: 10.1111/jofi.12855. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/jofi.12855 (visited on 10/28/2019).

Kosorok, Michael R. (2008). *Introduction to Empirical Processes and Semiparametric Inference*. Springer Series in Statistics. New York: Springer-Verlag. ISBN: 978-0-387-74977-8. URL: https://www.springer.com/gb/book/9780387749778 (visited on 04/10/2019).

Kuersteiner, Guido M. and Ingmar R. Prucha (Feb. 5, 2018). "Dynamic Spatial Panel Models: Networks, Common Shocks, and Sequential Exogeneity". In: arXiv: 1802.01755 [math, stat]. URL: http://arxiv.org/abs/1802.01755 (visited on 05/11/2020).

Lee, Jungyoon and Peter M. Robinson (Jan. 2016). "Series Estimation under Cross-Sectional Dependence". In: *Journal of Econometrics* 190.1, pp. 1–17. ISSN: 0304-4076. DOI: 10.1016/j.jeconom.2015.08.001.

Lewellen, Jonathan and Stefan Nagel (Nov. 1, 2006). "The Conditional CAPM Does Not Explain Asset-Pricing Anomalies". In: *Journal of Financial Economics* 82.2, pp. 289–314. ISSN: 0304-405X. DOI: 10.1016/j.jfineco.2005.05.012. URL: http://www.sciencedirect.com/science/article/pii/S0304405X06001371 (visited on 05/13/2020).

Li, Qi and Jeffrey Scott Racine (2007). *Nonparametric Econometrics: Theory and Practice*. Princeton University Press. Google Books: BIPiWazY0YC.

Lin, Zhongjian, Qi Li, and Yiguo Sun (Jan. 1, 2014). "A Consistent Nonparametric Test of Parametric Regression Functional Form in Fixed Effects Panel Data Models". In: *Journal of Econometrics*. Annals Issue: Misspecification Test Methods in Economet-

rics 178, pp. 167–179. ISSN: 0304-4076. DOI: 10.1016/j.jeconom.2013.08.014. URL: http://www.sciencedirect.com/science/article/pii/S0304407613001644 (visited on 04/26/2020).

Newey, Whitney (Sept. 1, 1985). "Generalized Method of Moments Specification Testing". In: *Journal of Econometrics* 29.3, pp. 229–256. ISSN: 0304-4076. DOI: 10.1016/0304-4076(85)90154-X. URL: http://www.sciencedirect.com/science/article/pii/030440768590154X (visited on 04/27/2020).

Newey, Whitney K. (1985). "Maximum Likelihood Specification Testing and Conditional Moment Tests". In: *Econometrica: Journal of the Econometric Society*, pp. 1047–1070.

Newey, Whitney K. and Daniel McFadden (Jan. 1994). "Chapter 36 Large Sample Estimation and Hypothesis Testing". In: *Handbook of Econometrics*. Vol. 4. Elsevier, pp. 2111–2245. DOI: 10.1016/S1573-4412(05)80005-4.

Robinson, P. M. (Nov. 3, 2011). "Asymptotic Theory for Nonparametric Regression with Spatial Data". In: *Journal of Econometrics*. Moment Restriction-Based Econometric Methods 165.1, pp. 5–19. ISSN: 0304-4076. DOI: 10.1016/j.jeconom.2011.05.002. URL: http://www.sciencedirect.com/science/article/pii/S0304407611000947 (visited on 05/29/2020).

Ross, Stephen A (Dec. 1976). "The Arbitrage Theory of Capital Asset Pricing". In: *Journal of Economic Theory* 13.3, pp. 341–360. ISSN: 0022-0531. DOI: 10.1016/0022-0531(76)90046-6.

Tauchen, George (Oct. 1985). "Diagnostic Testing and Evaluation of Maximum Likelihood Models". In: *Journal of Econometrics* 30.1-2, pp. 415–443. ISSN: 03044076. DOI: 10.1016/0304-4076(85)90149-6. URL: https://linkinghub.elsevier.com/retrieve/pii/0304407685901496 (visited on 04/27/2020).

Van der Vaart, Aad W. and Jon A. Wellner (1996). *Weak Convergence and Empirical Processes*. Springer Series in Statistics. New York, NY: Springer New York. ISBN: 978-1-4757-2547-6 978-1-4757-2545-2. DOI: 10.1007/978-1-4757-2545-2.

Wang, Kevin Q. (Mar. 1, 2002). "Nonparametric Tests of Conditional Mean-Variance Efficiency of a Benchmark Portfolio". In: *Journal of Empirical Finance* 9.2, pp. 133–169. ISSN: 0927-5398. DOI: 10.1016/S0927-5398(01)00049-4. URL: http://www.sciencedirect.com/science/article/pii/S0927539801000494 (visited on 03/27/2019).

Wang, Kevin Q. (2003). "Asset Pricing with Conditioning Information: A New Test". In: *The Journal of Finance* 58.1, pp. 161–196. ISSN: 1540-6261. DOI: 10.1111/1540-6261.00521. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1540-6261.00521 (visited on 03/20/2019).

Xiong, Wei (Nov. 2018). *The Mandarin Model of Growth.* w25296. Cambridge, MA: National Bureau of Economic Research, w25296. DOI: 10.3386/w25296. URL: http://www.nber.org/papers/w25296.pdf (visited on 05/19/2020).

Zu, Yang and H. Peter Boswijk (Mar. 1, 2017). "Consistent Nonparametric Specification Tests for Stochastic Volatility Models Based on the Return Distribution". In: *Journal of Empirical Finance* 41, pp. 53–75. ISSN: 0927-5398. DOI: 10.1016/j.jempfin.2016.12.005. URL: http://www.sciencedirect.com/science/article/pii/S092753981630161X (visited on 04/26/2020).